E-tivity 5: Network Analysis with Game of Thrones

1. Load Libraries & Data

Install and load the following libraries:

- tidyverse
- igraph
- ggraph

Load the dataset:

```
got_edges <- read_csv("GOT_network.csv")
```

2. Network Construction

Create the graph:

```
graph <- graph_from_data_frame(got_edges, directed = FALSE)
```

Simplify the graph:

```
graph <- simplify(graph, remove.multiple = TRUE, remove.loops = TRUE)
```

Calculate basic properties:

```
gorder(graph) # Number of characters
gsize(graph) # Number of interactions
edge_density(graph) # Network density
average.path.length(graph)
```

3. Degree Distribution

Calculate degree:

```
deg <- degree(graph)
```

Create a histogram using ggplot2 to visualize degree distribution.

Interpretation:

Characters with very high degree act as hubs and represent central figures in the story.

4. Centrality Measures

Calculate:

- Betweenness Centrality

- Closeness Centrality

- PageRank

Extract top 5 characters for each metric and compare their narrative importance.

5. Community Detection

Apply Walktrap algorithm:

communities <- cluster_walktrap(graph)

Calculate modularity:

modularity(communities)

Assign communities to nodes.

6. Visualization

Use ggraph with layout "kk":

- Size nodes by betweenness

- Color nodes by community

- Label only top 10 central characters

- Use low alpha for edges

Title:

Social Structure of the Seven Kingdoms

7. Assortativity

Calculate:

assortativity_degree(graph)

Interpretation:

Positive value = important characters connect with important ones.

Negative value = important characters connect with less connected ones.

Short Summary:

This analysis explores the social structure of the Game of Thrones universe by identifying hubs, bridge characters, and story communities using network theory and visualization techniques in R.